

02409 Multivariate Statistics - 2024

4 hours written exam - All aids allowed, including internet access.

There are 5 problems with a total of 30 questions.
The questions in a specific problem should be read in order.

The questions are weighted equally

A correct answer gives 5 points, a wrong answer gives -1 point. Unanswered questions or a “don’t know” give 0 points. The total number of points needed for a satisfactorily answered exam is determined at the final evaluation of the exam. Especially note, that the grade 10 may be given even if only one answer is wrong or unanswered.

If data is needed to solve a specific problem, it will be available in the problem description as an R and a SAS file.

The problem will specifically say, if you need to use the dataset.

R hint: rds files can be loaded as follows: `df <- readRDS('data.rds')`

PROBLEM 1 - Room Occupancy

We consider the dataset from this source: <https://archive.ics.uci.edu/dataset/864/room+occupancy+estimation>

The data we will use is available here in R (<https://designer.mcq.eksamen.dtu.dk/api/images/7b747e85-9052-4934-a6c8-74dfc605d75e>) and in SAS (<https://designer.mcq.eksamen.dtu.dk/api/images/69102c55-82f4-419b-a221-3da407ed532b>) .

The study aims at predicting room occupancy by means of a number of variables: Sound, temperature, CO2, etc.

We will only consider the sound and temperature variables, and thus use a subset of the variables

S1_Temp S2_Temp S3_Temp S4_Temp

S1_Sound S2_Sound S3_Sound S4_Sound

In this problem we will ignore the outcome *RoomCount*, i.e., how many people are present in the room.

Question 1.1

The covariance between S1_Temp and S3_Temp is:

Choose one answer

- ☐ 0.1424
- ☐ 0.0146
- ☐ 0.1234
- ☐ 0.7997
- ☐ 0.5863
- ☐ Don't know

Question 1.2

The 80% confidence interval for the *correlation* between S1_Sound and S2_Sound is:

Choose one answer

- ☐ [0.551; 0.569]
- ☐ [0.547; 0.573]
- ☐ Don't know
- ☐ [0.511; 0.548]
- ☐ [0.542; 0.577]
- ☐ [0.561; 0.595]

Question 1.3

Assuming the correlation stays unchanged, the minimum number of samples needed for the correlation between S1_Sound & S2_Sound - when tested against 0 - to have a p-value of less than 0.1 is:

Choose one answer

- ☐ 15
- ☐ 13
- ☐ 10
- ☐ 6
- ☐ 8
- ☐ Don't know

Question 1.4

Using a linear combination of the temperature variables, i.e. with no intercept, the fraction of variance explained of S1_Sound is:

Choose one answer

- ☐ Don't know
- ☐ 0.138
- ☐ 0.301
- ☐ 0.229
- ☐ 0.549
- ☐ 0.278

Question 1.5

We now assume that using a linear combination of the temperature variables, the fraction of variance explained of S4_Sound is 0.1530. When testing this value against 0, the resulting F-value is:

Choose one answer

- ☐ Don't know
- ☐ 367.42
- ☐ 457.19
- ☐ 585.36
- ☐ 418.27
- ☐ 113.83

Question 1.6

Using a linear combination of the temperature variables, the maximum fraction of variance that can be explained of a linear combination of the sound variables is:

Choose one answer

- ☐ Don't know
- ☐ 0.450
- ☐ 0.302
- ☐ 0.961
- ☐ 0.010
- ☐ 0.403

Problem 2 - Room occupancy, discrimination

We still consider the data from Problem 1, but now include the number of people present in the room: *RoomCount*

We now want to see if we can discriminate between the number of people present based on the Sound and Temp variables.

The occupancy of the room varies between 0 and 3, and we thus have four classes.

Question 2.1

Assuming equal priors and losses. The number of resubstitution misclassifications between class 0 and class 1 and vice versa - when going from Quadratic Discriminant Analysis to Linear Discriminant Analysis - increases by:

Choose one answer

- ☐ Don't know
- ☐ 315
- ☐ 673
- ☐ 299
- ☐ 412
- ☐ 29

Question 2.2

Assuming equal priors and losses. The overall resubstitution accuracy of Quadratic Discriminant Analysis and Linear Discriminant Analysis are: [QDA acc., LDA acc.]

Choose one answer

- ☐ [0.97, 0.93]
- ☐ [0.80, 0.74]
- ☐ [0.69, 0.68]
- ☐ [0.89, 0.81]
- ☐ [0.93, 0.88]
- ☐ Don't know

Question 2.3

Assuming equal priors and losses. Using LDA, we will now investigate whether the Sound variables - S1_Sound, ... , S4_Sound - provide no additional information to the discrimination between all four groups. The usual test statistic follows this distribution:

Choose one answer

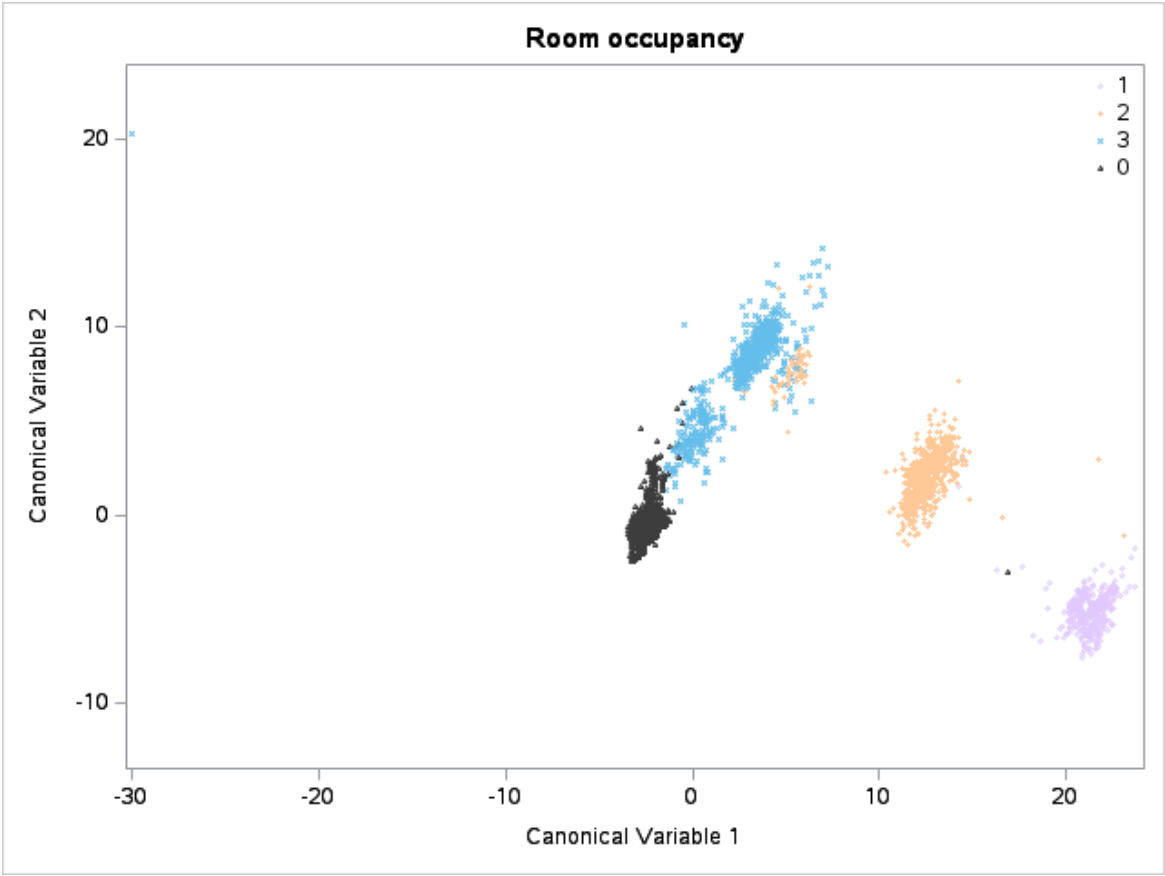
- ☐ F(4, 10120)
- ☐ F(8, 10121)
- ☐ U(8, 3, 10121)
- ☐ U(4, 4, 10129)
- ☐ $\chi^2(10121)$
- ☐ Don't know

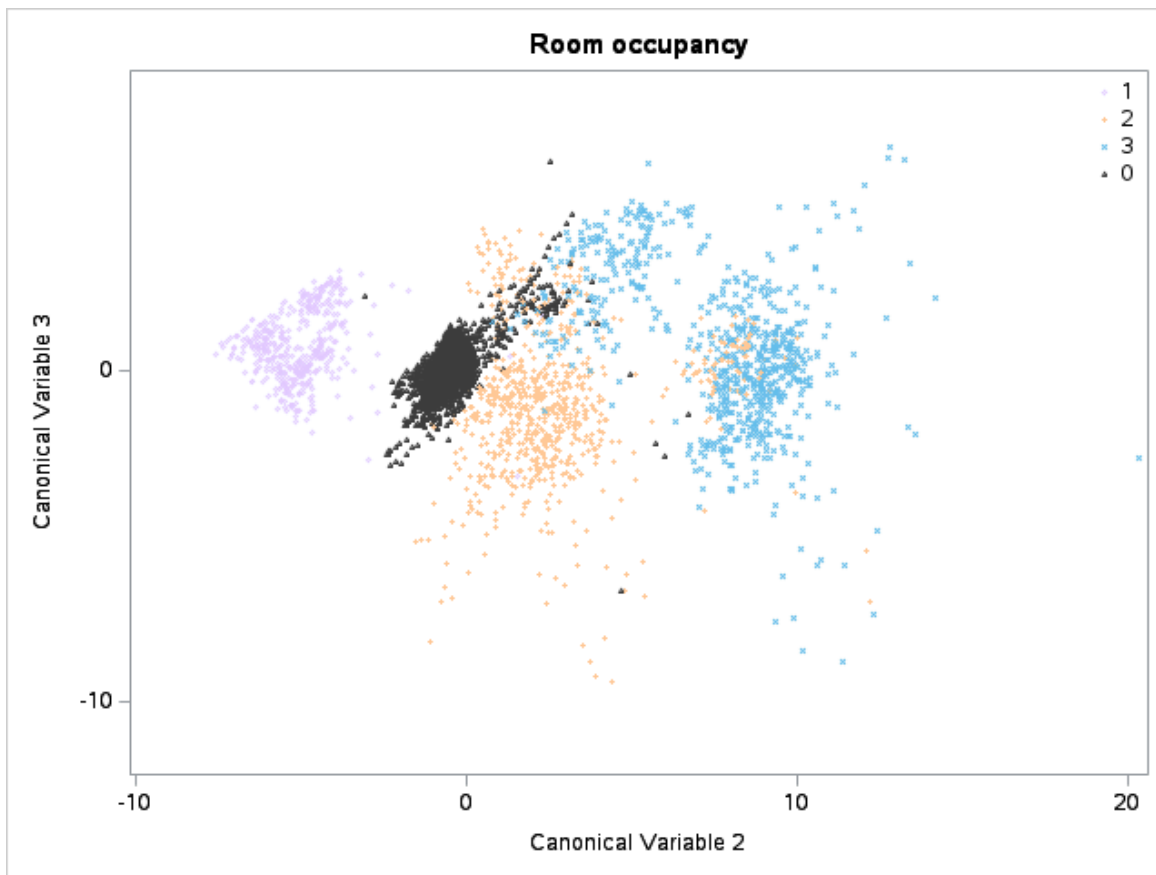
Question 2.4

Note: In this problem, you will be provided with the necessary outputs to answer the question.

We will now consider Canonical discrimination using all variables:

S1_Temp S2_Temp S3_Temp S4_Temp S1_Light S2_Light S3_Light S4_Light S1_Sound S2_Sound
S3_Sound S4_Sound S5_CO2 S5_CO2_Slope S6_PIR S7_PIR





Relying solely on the plots included in this question we can conclude:

Choose one answer

- ☐ The data are not perfectly separable with a linear classifier
- ☐ A perfectly separating hyperplane exists
- ☐ The data is perfectly separable using Quadratic Discriminant analysis.
- ☐ Don't know
- ☐ The data can be perfectly separated using the MiniMax principle
- ☐ Nothing, we need additional canonical variables shown before we can make conclusions

Problem 3 - Room Occupancy, Factor Analysis

We will continue our analysis of the Room occupancy dataset, now considering only the variables:

S3_Light S4_Light S5_CO2 S5_CO2_Slope S6_PIR S7_PIR

We will perform a factor analysis with VARIMAX rotation. The factors is estimated using the principal factor solution.

NOTE: In this problem you will be provided with the necessary outputs to answer each of the questions. The outputs will appear in the individual questions.

Question 3.1

How many factors must we include to account for 80% of the variance

	Eigenvalue	Difference	Proportion	Cumulative
1	3.06500280	2.10457684	0.5108	0.5108
2	0.96042596	0.07219669	0.1601	0.6709
3	0.88822926	0.32393981	0.1480	0.8189
4	0.56428945	0.18284511	0.0940	0.9130
5	0.38144434	0.24083616	0.0636	0.9766
6	0.14060819		0.0234	1.0000

Choose one answer

- ☐ 1
- ☐ 3
- ☐ Don't know
- ☐ 2
- ☐ 5
- ☐ 4

Question 3.2

	Eigenvalue	Difference	Proportion	Cumulative
1	3.06500280	2.10457684	0.5108	0.5108
2	0.96042596	0.07219669	0.1601	0.6709
3	0.88822926	0.32393981	0.1480	0.8189
4	0.56428945	0.18284511	0.0940	0.9130
5	0.38144434	0.24083616	0.0636	0.9766
6	0.14060819		0.0234	1.0000

We consider how many factors to retain. Regardless of the previous question, we consider retaining four factors. The test-statistic for the last two eigenvalues of the correlation matrix being equal, against all alternatives is:

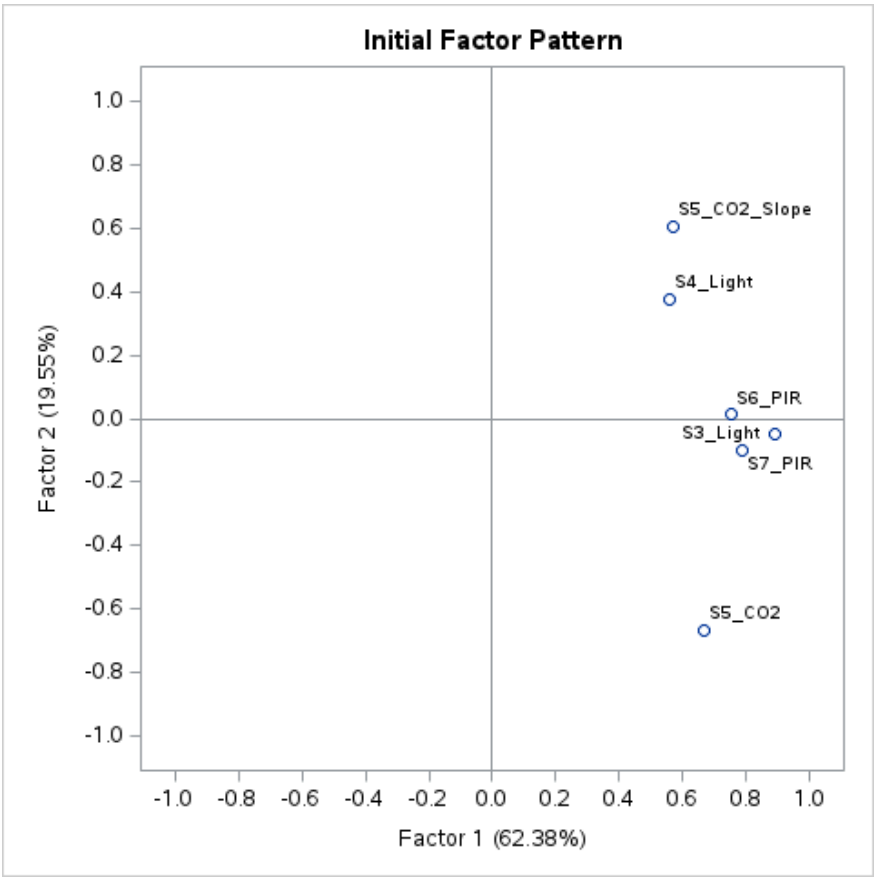
Choose one answer

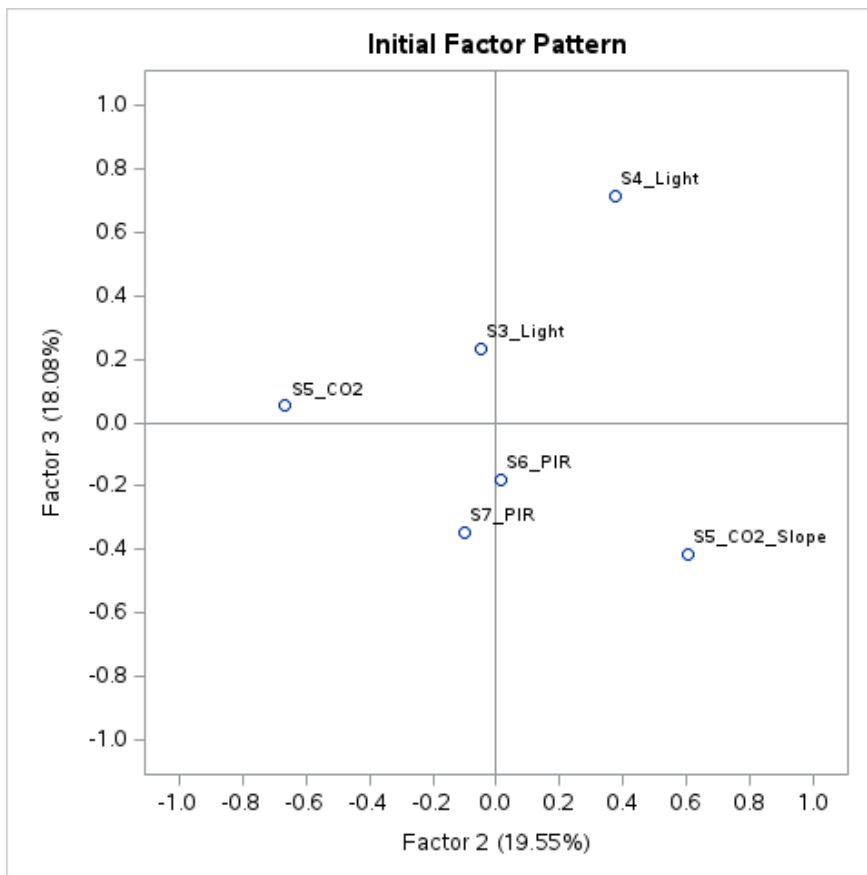
- ☐ 2423.87
- ☐ 23049.36
- ☐ 5322.25
- ☐ 4540.48
- ☐ 22124.40
- ☐ Don't know

Question 3.3

Initial Factor Pattern

	Factor1	Factor2	Factor3
S3_Light	0.89348	-0.04640	0.23224
S4_Light	0.55951	0.37313	0.71287
S5_CO2	0.66549	-0.66820	0.05554
S5_CO2_Slope	0.57004	0.60222	-0.41397
S6_PIR	0.75198	0.01530	-0.17995
S7_PIR	0.78763	-0.09830	-0.34537





We consider a factor analysis as described in the problem.
The initial factor 2 and factor 3 fits the following description:

Choose one answer

- ☐ Factor 2 is an average of S3_light, S6_PIR and S7_PIR. Factor3 is an average of S3_Light and S4_Light.
- ☐ Don't know
- ☐ Factor 2 is mainly a contrast of S5_CO2 against S4_Light and S5_CO2_Slope. Factor 3 is mainly an average of all the variables
- ☐ Factor 2 is a sum of S4_light and S5_CO2_Slope. Factor 3 is a contrast of S5_CO2 against S4_Light and S5_CO2_Slope.
- ☐ Factor 2 is mainly a contrast of S5_CO2 against S4_Light and S5_CO2_Slope. Factor 3 is mainly a contrast between S3_Light and S4_Light against S6_PIR, S7_PIR, and S5_CO2_Slope
- ☐ Factor 2 is mainly an average and Factor 3 is mainly S5_CO2_Slope

Question 3.4

Rotated Factor Pattern

	Factor1	Factor2	Factor3
S3_Light	0.64295	0.33783	0.57173
S4_Light	0.08579	0.11449	0.96953
S5_CO2	0.93899	-0.05035	0.09059
S5_CO2_Slope	-0.01442	0.91454	0.14963
S6_PIR	0.52105	0.53604	0.19813
S7_PIR	0.63560	0.58646	0.03727

Final Communality Estimates

S3_Light	S4_Light	S5_CO2	S5_CO2_Slope	S6_PIR	S7_PIR
0.85438728	0.96045827	0.89244227	0.85897641	0.59808453	0.74930926

We now consider the rotated factors. Factor2 and Factor3 describes the following fraction of the variance of S4_Light

Choose one answer

- ☐ 0.953
- ☐ 0.854
- ☐ 0.960/3
- ☐ $(0.11449+0.96953)/3$
- ☐ Don't know
- ☐ 0.960

Question 3.5

Initial Factor Pattern

	Factor1	Factor2	Factor3
S3_Light	0.89348	-0.04640	0.23224
S4_Light	0.55951	0.37313	0.71287
S5_CO2	0.66549	-0.66820	0.05554
S5_CO2_Slope	0.57004	0.60222	-0.41397
S6_PIR	0.75198	0.01530	-0.17995
S7_PIR	0.78763	-0.09830	-0.34537

The correlation matrix for the original variables is

	S3_Light	S4_Light	S5_CO2	S5_CO2_Slope	S6_PIR	S7_PIR
S3_Light	1	0.57948	0.65083	0.44771	0.50184	0.57781
S4_Light	0.57948	1	0.14861	0.21272	0.32454	0.22020
S5_CO2	0.65083	0.14861	1	0.06922	0.39526	0.47344
S5_CO2_Slope	0.44771	0.21272	0.06922	1	0.36837	0.42535
S6_PIR	0.50184	0.32454	0.39526	0.36837	1	0.57113
S7_PIR	0.57781	0.22020	0.47344	0.42535	0.57113	1

Note: If it is helpful for your calculations, you are welcome to use statistical software to recreate the correlation matrix.

We now consider the initial factors. The covariance between initial factor 1 and initial factor 2 is:

Choose one answer

- ☐ 0
- ☐ 5.02
- ☐ 3.30
- ☐ 6.53
- ☐ 9.39
- ☐ Don't know

Question 3.6

Initial Factor Pattern

	Factor1	Factor2	Factor3
S3_Light	0.89348	-0.04640	0.23224
S4_Light	0.55951	0.37313	0.71287
S5_CO2	0.66549	-0.66820	0.05554
S5_CO2_Slope	0.57004	0.60222	-0.41397
S6_PIR	0.75198	0.01530	-0.17995
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Rotated Factor Pattern

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S3_Light	0.64295	0.33783	0.57173
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S6_PIR	0.52105	0.53604	0.19813
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The correlation matrix for the original variables is

	S3_Light	S4_Light	S5_CO2	S5_CO2_Slope	S6_PIR	S7_PIR
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S7_PIR	0.57781	0.22020	0.47344	0.42535	0.57113	1

Note: If it is helpful for your calculations, you are welcome to use statistical software to recreate the correlation matrix.

We now consider both the initial and the rotated factors. The covariance between initial factor 1 and rotated factor 1 is:

Choose one answer

- ☐ 3.30
- ☐ 5.02
- ☐ 6.53

- ☐ 0
- ☐ 9.39
- ☐ Don't know

PROBLEM 4 - Toxicity

We consider the QSAR fish toxicity study from here: <https://archive.ics.uci.edu/dataset/504/qsar+fish+toxicity>

The data we will use is available here in R (<https://designer.mcq.eksamen.dtu.dk/api/images/55d44bce-fd41-406f-8af0-fbb939c13972>) and in SAS (<https://designer.mcq.eksamen.dtu.dk/api/images/06e393b2-da5e-4faf-a895-65fd5aaca89>).

The data set contains values for 6 attributes (molecular descriptors) of 908 chemicals used to predict quantitative acute aquatic toxicity towards the fish *Pimephales promelas* (fathead minnow). The outcome variable is LC50, the contraction at which 50 % of the fish die.

The original variable names are not important, and has been replaced as follows:

X1 = CIC0, X2 = SM1Dz, X3 = GATS1i, X4 = NdsCH, X5 = NdssC, X6 = MLOGP.

We will consider the model:

$$LC50 = \alpha + X1 + X2 + X3 + X4 + X5 + X6 + \epsilon$$

where α is an intercept and ϵ is the residuals

Question 4.1

The R^2 of the model is:

Choose one answer

- ☐ 0.9019
- ☐ 0.5772
- ☐ Don't know
- ☐ 0.3901
- ☐ 0.4135
- ☐ 0.5743

Question 4.2

The number of influential observations in the model, i.e., observations with both an absolute RSTUDENT larger than 2 and a leverage larger than 0.015 are:

Choose one answer

- ☐ 23
- ☐ 8
- ☐ 4
- ☐ Don't know
- ☐ 17
- ☐ 22

Question 4.3

We now consider the individual parameter estimates. Using DFBETAS, the following variables have observations with an absolute DFBETAS larger than 0.4

Choose one answer

- ☐ X2, X4, X5
- ☐ X1, X3, X6
- ☐ X2, X4, X6
- ☐ X4, X5
- ☐ X1, X4, X6
- ☐ Don't know

Question 4.4

We now consider multicollinearity. The following is true:

Choose one answer

- ☐ The TOL for X1 is less than 0.1 which indicates collinearity
- ☐ The TOL for X1 is larger than 10 which indicates collinearity
- ☐ The VIF for X6 is less than 0.1 which indicates collinearity
- ☐ There is no indication of multicollinearity
- ☐ The VIF for X6 is larger than 10 which indicates collinearity
- ☐ Don't know

Question 4.5

When performing a forward selection, the first variable to include is:

Choose one answer

- ☐ X5
- ☐ X1
- ☐ X3
- ☐ Don't know
- ☐ X6
- ☐ X4

Question 4.6

We want to estimate a 95%-confidence interval for the expected value of observation 1. Given new additional information we have been informed that the variance can be assumed to be $V(U) = \sigma^2 \cdot 0.01$. The 95%-confidence interval is then:

Choose one answer

- ☐ [3.74, 3.96]
- ☐ [3.77, 3.93]
- ☐ Don't know
- ☐ [3.58, 3.96]
- ☐ [3.70, 4.01]
- ☐ [3.66, 4.04]

Question 4.7

Regardless of the previous question we now consider observation 6.

We want a prediction interval for a new observation and due to degradation of sensors the variance has increased and the new observation Y_q follows $Y_q \sim N(E(Y), 2 \cdot \sigma^2)$. The 95% prediction interval is

Choose one answer

- ☐ [-0.16, 5.12]
- ☐ Don't know
- ☐ [0.27, 4.70]
- ☐ [2.32, 2.64]
- ☐ [0.61, 4.35]
- ☐ [-0.82, 4.46]

Question 4.8

We now propose a new model where we remove X_1 and X_6 . When we test the original model against the new proposed model the resulting F-value is:

Choose one answer

- ☐ 247.29
- ☐ Don't know
- ☐ 160.01
- ☐ 118.94
- ☐ 204.97
- ☐ 0.3568

Problem 5

We consider the following model

$$\begin{bmatrix} Y_1 & Z_1 \\ Y_2 & Z_2 \\ Y_3 & Z_3 \\ Y_4 & Z_4 \\ Y_5 & Z_5 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} \alpha_y & \alpha_z \\ \beta_y & \beta_z \\ \gamma_y & \gamma_z \end{bmatrix} + \begin{bmatrix} \delta_1 & \varepsilon_1 \\ \delta_2 & \varepsilon_2 \\ \delta_3 & \varepsilon_3 \\ \delta_4 & \varepsilon_4 \\ \delta_5 & \varepsilon_5 \end{bmatrix}$$

where the error terms $[\delta_i \ \varepsilon_i]$ for $i=1,\dots,5$ are independent and normally distributed $N_2(\mathbf{0}, \mathbf{\Sigma})$, and where $\mathbf{\Sigma}$ is the unknown dispersion matrix.

We have obtained the following observations

$$\begin{bmatrix} 21 & 17 \\ 16 & 12 \\ 17 & 13 \\ 19 & 16 \\ 25 & 19 \end{bmatrix}$$

and the unbiased estimate of $\mathbf{\Sigma}$ is

$$\begin{bmatrix} 1.13333333 & 0.36666667 \\ 0.36666667 & 0.13333333 \end{bmatrix}$$

Question 5.1

The maximum likelihood estimate for the parameters $[\alpha_z \quad \beta_z \quad \gamma_z]$ are:

Choose one answer

- ☐ [11.33, 3.87, 6.47]
- ☐ Don't know
- ☐ [7.7, 4.1, 5.5]
- ☐ [0.31, -0.18, -0.18]
- ☐ [2.64, -1.51, -1.51]
- ☐ [9.67, 5.51, 4.47]

Question 5.2

The estimated covariance between α_y and α_z is:

Choose one answer

- ☐ 0.86
- ☐ -1.51
- ☐ Don't know
- ☐ 0.37
- ☐ 0.31
- ☐ 1.13

Question 5.3

The variance of the maximum likelihood estimate of β_y is:

Choose one answer

- ☐ 0.311
- ☐ 0.124
- ☐ 1.133
- ☐ 1.058
- ☐ 2.644
- ☐ Don't know

Question 5.4

We now want to test if α_y is equal to 11 against all alternatives. The P-value is:

Choose one answer

- ☐ 0.020
- ☐ Don't know
- ☐ 0.428
- ☐ 0.857
- ☐ 0.001
- ☐ 0.987

Question 5.5

We now want to test if $[\beta_y \quad \beta_z] = [4 \quad 4]$ against all alternatives. The test statistic for this test has - under the null-hypothesis the following distribution:

Choose one answer

- ☐ U(2, 2, 2)
- ☐ U(1, 2, 2)
- ☐ U(2, 1, 3)
- ☐ U(2, 1, 2)
- ☐ U(1, 2, 3)
- ☐ Don't know

Question 5.6

We now want to test if $[\alpha_y \quad \beta_y \quad \gamma_y] = [11 \quad 4 \quad 6]$ against all alternatives. Wilks Lambda for this test is:

Choose one answer

- ☐ 0.453
- ☐ 0.222
- ☐ Don't know
- ☐ 0.004
- ☐ 0.586
- ☐ 0.001

